

Formalization of Gödel’s Incompleteness Theorems in Basic Recursive Arithmetic (BRA)

Summary of an Agda development

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Abstract

We present a fully formalized proof of both Gödel’s First and Second Incompleteness Theorems in a tree-based variant of Guard’s Basic Recursive Arithmetic (BRA), executed in Agda 2.8 with `--safe --without-K --exact-split`. The development contains **zero postulates and zero holes**; every per-file solo typecheck completes in under one second (with one historical exception, the 16k-line dispatcher cascade for *thmT*).

The headline results are:

- *thm11* : $Deriv\ G \rightarrow Deriv\ \perp$ (Gödel I: provability of the diagonal sentence implies $\perp$);
- *godelII* : $Deriv\ Con \rightarrow Deriv\ \perp$ (Gödel II: provability of consistency implies $\perp$).

The development follows Guard’s 1963 lecture notes on recursive arithmetic exclusively. The intermediate construction is a single first-order computable function *constr5* : *Fun1* realizing Guard’s “*F(x)*” (page 17 of his notes); each step in the chain is given by an explicit equational derivation in BRA.

1 Introduction

The aim of the project is to give a feasible, ASCII-only, syntax-first Agda formalization of Gödel’s two incompleteness theorems, following Guard’s 1963 system of *Basic Recursive Arithmetic* [1]. The intent is to avoid historical baggage (Gödel numbering of Hilbert proofs, primitive recursive arithmetic with explicit numerals, “encoded” substitution as a Hilbert-system meta theorem) by working inside a small tree-based equational system from the outset:

- Terms are built from *O* (leaf), *var n*, *ap₁ f t*, *ap₂ g t₁ t₂*.
- Function symbols come in two arities: *Fun₁* and *Fun₂*, with primitive constants $\{I, Fst, Snd, Comp, Comp_2, Z\}$ and $\{Pair, Const, Lift, Post, Fan, IfLf, TreeEq, treeRec\}$.
- Equations are objects *eqn* ℓ *r* between terms; formulas *Formula* extend equations with *not* and *imp*.
- Provability is an inductive type *Deriv F* with constructors for the algebraic axioms (*ax_{*}*), the equality axioms, the propositional axioms (*ax_K*, *ax_S*, *ax_{Neg}*, *ax_{ExFalso}*, *ax_{Contrapos}*), modus ponens (*mp*), substitution (*ruleInst*), and binary tree induction (*ruleIndBT*).

The natural numbers are represented as binary trees (*lf* for 0, *nd lf n* for the successor in unary). The encoding *reify* : *Tree* $\rightarrow$ *Term* maps *lf* $\mapsto O$ and *nd a b* $\mapsto ap_2\ Pair\ (reify\ a)\ (reify\ b)$.

2 The provability predicate $thmT$

The Gödel-style “ th ” provability predicate is realized in BRA by a single Fun_1 -functor:

$$thmT = Rec\ stepProtoWrapped : Fun_1.$$

Here $stepProtoWrapped$ is a Fun_2 assembled from a 40-deep linear $IfLf$ cascade (BRA/Thm/StepProto.agda and BRA/Thm/ThmT.agda, ~16k lines), one branch per primitive proof rule: ten equational axioms of Group I (axI, axFst, ..., axKT), eight Group II axioms (axIfLfL, axIfLfN, four axTreeEq*, axGoodstein), nine equality and propositional rules, and three rules of inference (mp , $ruleInst$, $ruleIndBT$).

The *key* property of $thmT$ is that on the encoded form of any proof tree it returns the encoded conclusion of that proof. This is the content of the *encoding-soundness* lemma $encode : (P : Formula) \rightarrow Deriv\ P \rightarrow \Sigma\ Tree\ (\lambda y \rightarrow Deriv\ (thmT\ (reify\ y) = reify\ (codeFormula\ P)))$, proved by induction on the $Deriv$ derivation in BRA/Thm/ThmT.agda.

2.1 Self-substitution closure

A second key property is that $thmT$ contains no free variable (it is a closed combinator):

$$thClosed : \forall x\ r, \quad substF_1\ x\ r\ thmT = thmT$$

and that the encoding of $thmT$ is var-free:

$$codeF1Th_noVar : \forall x\ repl, \quad codedSubst\ repl\ (code\ (var\ x))\ (codeF_1\ thmT) = codeF_1\ thmT.$$

Both are `refl` inside the abstract block where $thmT$ unfolds.

3 Gödel’s First Incompleteness Theorem (Theorem 11)

3.1 The diagonal sentence

Following Guard, define

$$\begin{aligned} F_{pre} &:= \neg(thmT\ (v_1) = sub\ (v_0)\ (v_0)), \\ j &:= codeFormula\ F_{pre} \quad (\text{a closed Tree}), \\ G &:= F_{pre}[v_0 := reify\ j] = \neg(thmT\ (v_1) = sub\ i\ i), \end{aligned}$$

where $i := reify\ j$ and sub is the BRA-internal coded substitution function (BRA/Sub.agda). The sentence G has v_1 free; informally, G asserts of itself that no proof tree v_1 proves the formula whose code is $j[v_0 := i]$.

3.2 Theorem 11

Pulling apart the abstract closure (BRA/Thm11Abstract.agda), the precise statement is:

Theorem 1 (thm11, Gödel I). $thm11 : Deriv\ G \rightarrow Deriv\ \perp$, where $\perp := atomic\ (eqn\ trueT\ falseT)$ with $trueT := O$ and $falseT := ap_2\ Pair\ O\ O$.

Sketch. From any $\pi : Deriv\ G$ extract its encoding (y_d, E_1) where $E_1 : Deriv\ (thmT\ (reify\ y_d) = reify\ (codeFormula\ G))$. A separate *diagonal lemma* **diagonal** (BRA/Thm11Diagonal.agda) yields $N : Deriv\ \neg(thmT\ (reify\ y_d) = reify\ (codeFormula\ G))$ from π , and $ax_{ExFalse} +$ two mp ’s collapse E_1 and N to $Deriv\ \perp$. $\square$

The diagonal lemma is constructed inside an abstract block in `BRA/Thm11Diagonal.agda` using `thClosed`, `codeF1Th_noVar`, and `cc` (the local instance of the meta-equation `codedSubst (code i) (code v0) j = codeFormula G`, which is provable by induction on F_{pre} via `codeCommutesFormula` in `BRA/CodeCommutes.agda`).

4 Gödel’s Second Incompleteness Theorem (Theorem 14)

The second theorem is the more substantial construction. It says that the consistency formula *Con* entails $\perp$:

Theorem 2 (`godelIII`, Gödel II). *godelIII* : $\text{Deriv } \text{Con} \rightarrow \text{Deriv } \perp$, where $\text{Con} := \neg(\text{thmT } (v_0) = \text{reify } (\text{codeFormula } \perp))$ (consistency: “no proof, encoded as v_0 , derives $\perp$ ”).

The construction follows Guard’s pages 16–17 step by step.

4.1 Numerals: the asymmetry $f(\underline{x})$ vs. $\underline{f(x)}$

A crucial feature of the Theorem 12–14 chain, easy to miss in the formalisation, is the *asymmetric* role of the numeral function. In Guard’s notation [1] the underline $\underline{x}$ denotes the *numeral* of x : a closed BRA term whose value is x . In our Agda development this is the primitive-recursive function

$$\text{num} = \text{cor} : \text{Tree} \rightarrow \text{Tree},$$

which sends a tree x to the code of the BRA term that *names* x . We use *num* and *cor* interchangeably with Guard’s $\underline{x}$.

The asymmetry is this: for any Fun_1 -functor f and tree x ,

- $\text{ap}_1 f (\text{num } x)$, i.e. $f(\underline{x})$, is the code of a BRA term — specifically the term “ f applied to the numeral of x ”;
- $\text{num } (f x)$, i.e. $\underline{f(x)}$, is *generally not* the code of any such syntactic term — it is the code of the numeral naming the *value* $f x$.

The two trees disagree in general; they agree only after BRA proves them equal.

Theorem 12 is precisely the internalisation of that equation: for every $f : \text{Fun}_1$ it builds a $D_f : \text{Fun}_1$ such that

$$\text{Deriv } (\text{thmT } (D_f x) = \text{codeFTeq}_1 f x),$$

where the right-hand side $\text{codeFTeq}_1 f x$ is the encoded equation $\text{ap}_1 f (\text{cor } x) = \text{cor } (f x)$, i.e. the encoded form of $f(\underline{x}) = \underline{f(x)}$. In other words: BRA itself proves the bridge between the syntactic side (left-hand: a code-of-term) and the semantic side (right-hand: the numeral of the value). This is what licenses every subsequent step of Goedel II: it lets the internal *thmT* predicate move between the two encodings exactly where Guard’s pencil-and-paper argument tacitly does so.

A symmetric variant in which both sides are coded the same way (*code* $(f x)$ on both sides, say) collapses: the resulting trees do not align across the five steps of Theorem 14, and step 5 becomes underivable. The asymmetric form $f(\underline{x}) = \underline{f(x)}$ is therefore not a presentational choice but a load-bearing feature of the proof.

4.2 Theorem 12 — the encoded “ $th\ f$ ” clause

For every Fun_1 -functor f , Theorem 12 produces a $Fun_1\ D_f$ such that under the canonical encoding, $thmT$ evaluated at $D_f\ x$ is provably equal to the encoded form of “ $f(num\ x) = num\ (f\ x)$ ”. Formally:

Theorem 3 ($thm12_F_1$, parametric Theorem 12 for Fun_1). $thm12_F_1 : (f : Fun_1) \rightarrow Thm12_F_1_Result\ f$ where the record $Thm12_F_1_Result\ f$ packages a $D_f : Fun_1$ together with $proof : (x : Term) \rightarrow Deriv\ (thmT\ (D_f\ x) = codeFTeq_1\ f\ x)$. The conclusion $codeFTeq_1\ f\ x$ is the literal Pair-shape encoded equation $ap_1\ f\ (cor\ x) = cor\ (f\ x)$.

(Analogous statement for Fun_2 .) This is a structural recursion on f : there is one clause per Fun_1 / Fun_2 constructor (15 total), each in its own `BRA/Thm12/Parts/*.agda` file. The composite cases (*Comp*, *Comp₂*, *Lift*, *Post*, *Fan*, *treeRec*) dispatch to per-case modules; the recursive *TreeEq* case uses `ruleIndBT2` (2-tree-pair induction with diagonal cross-IHs) to establish closure. The whole theorem is in `BRA/Thm12.agda`, module `FromBridges`, and contains zero postulates / zero holes.

4.3 Theorem 13 — under hypothesis “ $f\ x = y$ ”

Building on Theorem 12, one produces under the hypothesis $f\ x = y$ the encoded form of “ $f(num\ x) = num\ y$ ”:

Lemma 4 ($thm13_F_1$). For every $f : Fun_1$ and every Theorem 12 result r_{12} for f , $thm13_F_1\ f\ r_{12}\ x\ y : Deriv\ (f\ x = y) \rightarrow Deriv\ (thmT\ (D_f\ x) = codeFTeq_{1,Hyp}\ f\ x\ y)$ where the conclusion replaces the “ $cor\ (f\ x)$ ” slot of $codeFTeq_1$ by $cor\ y$. One line: `ruleTrans` on $r_{12}.proof\ x$ with `congR Pair` on $cong_1\ cor$ of the hypothesis.

4.4 Theorem 14 — the consistency-to-bot construction

This is the heart of Gödel II. The contract abstracted in `BRA/Thm14Abstract.agda` asks for:

1. a Fun_1 -functor $constr5$ such that $thmT\ (constr5\ x)$ syntactically encodes $\perp$;
2. a parametric internal-implication

$step5 : (x : Term) \rightarrow Deriv\ ((thmT\ x = reify\ (codeFormula\ G))\ imp\ (thmT\ (constr5\ x) = reify\ (codeFormula\ G)))$

The closure $closureToG$ then applies $ax_{Contrapos}$ to $step5$ instantiated at v_1 , $ruleInsts\ Con$ at $constr5\ v_1$, modus-ponenses to obtain $\neg(thmT\ v_1 = i)$, transports through $subIteqG\ (sub\ i\ i = cor\ G)$ and G_{unfold} (propositional normal form of G), arriving at $Deriv\ G$. Composing with $thm11$ gives $godelII : Deriv\ Con \rightarrow Deriv\ \perp$.

4.4.1 Construction of $constr5$

Following Guard’s page 17, the construction relies on two pre-existing tautologies, encoded:

- $t := (v_0 = v_2)\ imp\ ((v_1 = v_2)\ imp\ (v_0 = v_1))$ (transitivity);
- $t' := (v_0 = v_1)\ imp\ ((\neg(v_0 = v_1))\ imp\ \perp)$ (ex falso).

Both are derivable in BRA (BRA/Thm14Numerals.agda), and via *encode* they yield closed Trees *tterm*, *t'term* such that *thmT* (*reify tterm*) = *reify (codeFormula t)* (and analogously for *t'*).

The chain is:

$$\begin{aligned}
f_part(x) &:= ruleInst\ v_0\ (th\ \hat{x})\ (ruleInst\ v_1\ (sub\ i\ i)\ (ruleInst\ v_2\ cor\ G\ tterm)), \\
g_part(x) &:= mp(mp(f_part(x), D_{thmT}\ x), D_{sub}\ i\ i), \\
K_part(x) &:= ruleInst\ v_1\ (cor\ x)\ x, \\
h_part_skel(x) &:= ruleInst\ v_0\ (th\ \hat{x})\ (ruleInst\ v_1\ (sub\ i\ i)\ t'term), \\
constr5_final(x) &:= mp(mp(h_part_skel(x), g_part(x)), K_part(x)).
\end{aligned}$$

The substituents are the encoded mixed-form Pairs of Guard: $\widehat{th\ \hat{x}} := ap_2\ Pair\ (reify\ tagAp_1)\ (ap_2\ Pair\ (reify\ (code\ sub\ i\ i)))$, and $cor\ G := reify\ (code\ (reify\ (codeFormula\ G)))$.

4.4.2 The five steps and their lemmas

Following Guard's enumeration (guard15.pdf pages 16–17):

Step 1 — *step1_l* (Th14Constr5.agda). By Theorem 13 applied to *thmT* at $(x, cor\ G)$ under hypothesis *thmT* $x = cor\ G$, get *Deriv* (*PrAtX* $x\ imp\ thmT\ (D_{thmT}\ x) = codeFTeq_{1,Hyp}\ thmT\ x\ cor\ G$).

Step 2 — *step2_l* (Th14Step2.agda). By Theorem 13 applied to *sub* at $(i, i, cor\ G)$ under closed hypothesis *subIleqcG*.

Step 3 — *step3_l* (= *g_part_l*) (Th14Step3.agda, Th14Step3Canon.agda). Three sb-applications on *t* via *thmTSubLemma*, canonicalized via *subT_codeFormula_imp/atomic* and leaf preservation lemmas, then two *mp* dispatches at *g_part_inner* and *g_part*. Output: *Deriv* (*PrAtX* $x\ imp\ thmT\ (g_part\ x) = th\ \widehat{\hat{x} = sub\ i\ i}$).

Step 4 — *K_part_l* (Th14Step4Final.agda). One sb-application at v_1 on the diagonal sentence G_{norm} via *thmTSubLemma* + the *PrAtX* x hypothesis to swap *thmT* $x \mapsto cor\ G$, canonicalized via *subT_codeFormula_neg/atomic* and leaf preservation. Output: *Deriv* (*PrAtX* $x\ imp\ thmT\ (K_part\ x) = th\ \widehat{\hat{x} \neq sub\ i\ i}$).

Step 5 — *step5_l* (Th14Step5.agda). Two outer *mp* dispatches at *constr5_inner*, *final* and *constr5_final*, with *step3_l* feeding the inner antecedent and *K_part_l* standing in syntactically as the second-*mp* witness. The dispatcher's syntactic projection of the consequent gives the encoded $\perp$. Output: *Deriv* (*PrAtX* $x\ imp\ thmT\ (constr5_final\ x) = reify\ (codeFormula\ \perp)$).

Auxiliary — *h_part_skel_l* (Th14Step5HPart.agda). Two sb-applications on *t'*, mirror of step 3 (one fewer layer because *t'* has only two free variables). Output: *thmT* (*h_part_skel* x) = *ex falso* form. Used as *step5_l*'s inner *d1*.

The headline composition (BRA/GoedelIIIFull.agda) is then:

```

r12_th  : Thm12_F1_Result thmT
r12_th  = FromBridges.thm12_F1 thmT

r12_sub : Thm12_F2_Result sub
r12_sub = FromBridges.thm12_F2 sub

```

```

module Step5 = BRA.Th14Step5.Th14Step5 r12_th r12_sub
module Final = BRA.GoedelIII.Compose Step5.constr5_final Step5.step5_l

```

```

godelIII : Deriv Con -> Deriv bot
godelIII = Final.godelIII

```

4.5 Modal-logic perspective: Theorems 12–14 as $\Box$ -functoriality and HBL

The asymmetry of Section 4.1 and the structure of Theorems 12–14 admit a clean reading in intuitionistic modal $S4$. Following Davies and Pfenning [3], separate *value variables* (in Γ) from *code variables* (in Δ) and write $\Box A$ for “code of type A ”, with introduction **box** $E : \Box A$ when $\Delta; \cdot \vdash E : A$ (only modal-variable dependencies allowed) and elimination **let box** $u = E_1$ **in** E_2 . This two-context discipline is precisely the truth/validity distinction.

Read in this light:

The *cor/code* asymmetry is the truth/validity distinction. Guard’s $\underline{x}$ is the modal coercion sending a closed value x to its code (a **box** at type $\Box A$); *code* of an open BRA term has free value variables and lives outside the modal fragment. The trap that “ $\underline{f(x)}$ is the code of $f(x)$ ” is precisely the trap of confusing **box** $(f\ x)$ with f (**box** x) in $\lambda_e^{\rightarrow\Box}$, where the typing rules forbid the confusion outright.

Theorem 12 is the functoriality of $\Box$ over Fun_1 / Fun_2 . At the BRA level a Fun_1 -functor f is a primitive recursive map $A \rightarrow B$ (here $A = B = Tree$). Theorem 12 produces a $D_f : Fun_1$ realising the lifted map $\Box A \rightarrow \Box B$, witnessed by an internal derivation of the naturality square

$$thmT (D_f\ x) = \widehat{f(\underline{x})} = \underline{f(x)}.$$

The two sides of the encoded equation are the two paths $x \mapsto f(\text{cor } x)$ and $x \mapsto \text{cor}(f\ x)$ in the unit-naturality square of $\text{cor} : \text{id} \Rightarrow \Box$. Theorem 12 is the BRA-internal proof that these two paths agree at every f .

Theorem 13 transports an equation across $\Box$, a derived rule of the same flavour as the modal K axiom $\Box(A \rightarrow B) \rightarrow \Box A \rightarrow \Box B$. Given a value-level *Deriv* $(f\ x = y)$ it produces an internal derivation of $thmT (D_f\ x) = \widehat{f(\underline{x})} = y$, replacing $f(x)$ by y inside the box. Operationally this is necessitation-under-hypothesis: an equation that holds outside the modality is reflected to an equation inside it.

Theorem 14, step 5, is an HBL-style internal reflection. *step5_l* derives, parametrically in x , the encoded internal implication

$$thmT\ x = \text{cor } G \quad \text{imp} \quad thmT\ (\text{constr5 } x) = \text{cor } \perp.$$

This is the Hilbert–Bernays–Löb clause D2 ($\text{Pr} \ulcorner P \urcorner \rightarrow \text{Pr} \ulcorner \text{Pr} \ulcorner P \urcorner \urcorner$, “provable propositions are provably provable”) applied at $P := G$, combined with G ’s own self-referential content $G \leftrightarrow \neg \text{Pr}(\ulcorner G \urcorner)$ to yield $\Box \perp$. *constr5* is the explicit primitive recursive realiser of D2 at this fixed point; the chain *f_part*, *g_part*, *K_part*, *h_part_skel* assembled in Section 4.1 above is the syntactic decomposition of that realiser.

What this perspective does *not* do is shorten the BRA proof. Davies and Pfenning’s $\Box$ is a metatheoretic construct; their typing rules leave $\Box A$ abstract and give no algorithmic content for D_f , $constr5$, or $thmT$. Gödel II proper requires the provability predicate to be a primitive recursive function of the object theory, definable inside BRA itself — the deep encoding is intrinsic to the theorem, not an artefact of the formalisation. The modal reading is therefore best understood as a *specification* that the BRA construction realises: theorem by theorem, BRA delivers the content that an *S4*-style code modality would require, via an explicit primitive-recursive witness.

5 Methodology

5.1 Fast typecheck as a correctness oracle

Every per-file solo typecheck runs in well under one second (with the historical exception of BRA/Thm/ThmT.agda at ~ 19 s; this is the 16k-line dispatcher cascade and was already in place before the Gödel II push). The session followed a strict rule: *slow typecheck = mathematical mismatch, stop and re-architect*. In practice this prevented two specific traps:

1. Trying to feed Agda an opaque normalisation it cannot perform (e.g. unfolding *codeFormula G* at a parametric substituent position). The fix is always: add a single targeted refl-witnessable lemma inside the right **abstract** block (where the opaque definition unfolds), and use it as a black box outside.
2. Forcing structural recursion through opaque trees (*codedSubst/codedSubstT* on *codeF₁ thmT*, etc.). Same fix: prove the equation once inside the abstract block, expose only the type signature outside.

5.2 Architectural choice for *constr5*

The original Th14Constr5Final.constr5 wraps *h_part* with two *additional* encoded-mp layers (*h_part_thxLayer*, *h_part*). Under the *thmT* dispatcher these would consume both antecedents of *h_part_skel*’s ex-falso shape, collapsing the value of *thmT* (*h_part x*) to $\perp$ before it can serve as antecedent to *constr5_inner*. The fix is to use *h_part_skel* directly in *constr5_{final}* (pure sb-chain, no internal mp’s), matching Guard’s intent on page 17 and the comment in Th14Constr5.agda. The two outer *mp*’s in *constr5_{final}* then consume the two antecedents exactly as Guard’s chain prescribes.

6 Statistics

6.1 File-by-file summary of the Gödel II push

File	LoC	Solo typecheck
BRA/Th14SubTLeaf.agda	374	0.18 s
BRA/Th14Step3Canon.agda	582	0.60 s
BRA/Th14Step3.agda	633	0.81 s
BRA/Th14Step4Final.agda	213	0.84 s
BRA/Th14Step5HPart.agda	394	0.84 s
BRA/Th14Step5.agda	305	0.76 s
BRA/GoedelIIFull.agda	50	0.71 s

6.2 Earlier infrastructure (already in place)

- BRA/Thm/ThmT.agda: 16k LoC, 40-deep dispatcher cascade, 19 s.
- BRA/Thm12.agda: structural Theorem 12 with one Parts file per constructor, 0.65 s.
- BRA/Thm13.agda: ~ 80 LoC, 0.1 s.
- BRA/Thm11.agda, Thm11Abstract, Thm11Diagonal: ~ 300 LoC total.
- BRA/Thm14Abstract.agda: ~ 250 LoC, the *closureToG* machinery.
- BRA/Thm14Constr5Final.agda, Th14StepHPre, Th14HUnfolds, Th14GPartUnfolds: ~ 700 LoC of Pair-of-Comp2 unfold lemmas.

6.3 Constraints maintained throughout

- `--safe --without-K --exact-split` on every file.
- ASCII only. No Unicode.
- No postulates. No holes. No `with`-abstraction. No dot patterns.
- No new *Deriv* axioms beyond Guard’s primitive list.

7 Discussion

The two Gödel theorems in BRA, side by side:

- **Gödel I.** If the diagonal sentence G is provable, then $\perp$ is provable. Equivalently (under the meta-assumption that BRA is consistent): G is unprovable.
- **Gödel II.** If the consistency formula Con is provable, then $\perp$ is provable. Equivalently: BRA cannot prove its own consistency.

The intermediate functions $constr5_{\text{final}}$ and the parametric step lemmas $step1_l\text{--}step5_l$, K_part_l , $h_part_skel_l$ are *first-order* in the sense of recursive-arithmetic definability: each is a closed combinator built from primitive Fun_1 / Fun_2 symbols, and each *Deriv* derivation is given by an explicit equational proof in the BRA system. No external soundness theorem is invoked.

7.1 Comparison with prior formalisations

To our knowledge the only previous machine formalisation of Gödel’s *Second* Incompleteness Theorem is Paulson’s Isabelle/HOL development of hereditarily finite set theory [2]. Other formalisations (O’Connor in Coq, Han in Lean, Carneiro’s metamath work) target Gödel’s *First* theorem only.

Paulson’s development uses HF set theory and full Gödel numbering of Hilbert-style proofs, with encoded substitution treated as an external soundness theorem. The present BRA development by contrast follows Guard [1]: the “provability predicate” $thmT$ is itself a primitive-recursive combinator built from $Rec + Fan + Lift + IfLf$; encoded substitution is performed by an explicit primitive recursive function $sub : Fun_2$; and the entire chain (Gödel I and II) stays first-order, no Hilbert-proof Gödel numbering, no external soundness theorem.

References

- [1] J. R. Guard, *Lecture notes on recursive arithmetic*, Applied Mathematics Division, Argonne National Laboratory, 1963.
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- [3] R. Davies and F. Pfenning, *A modal analysis of staged computation*, Journal of the ACM, vol. 48 no. 3, May 2001, pp. 555–604.